

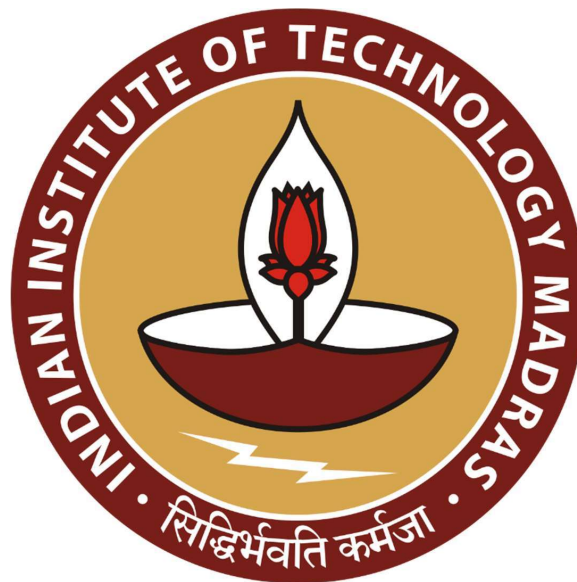
A Business Data Management Report on Sales Insights and Operational Challenges in a Local Bangle and Wool Store

A Mid Term report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “**A Business Data Management Report on Sales Insights and Operational Challenges in a Local Bangle and Wool Store**”. I extend my appreciation to **Naina Kangan and Wool Store**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:

A handwritten signature in black ink that reads "Vivek Mittal". The signature is written in a cursive style with a large initial 'V'.

Name: Vivek Mittal

Date: January 8, 2026

1 Executive Summary and Title

Project Title: “A Business Data Management Report on Sales Insights and Operational Challenges in a Local Bangle and Wool Store”

This report analyzes sales data and performance of Naina Kangan and Wool Store, a local bangles and wool business located at New Market, Doiwala, Dehradun (248140), Uttarakhand, serving wholesale bangle clients and retail wool buyers.

The study aims to understand sales patterns, profitability, seasonality and customer behavior using transaction-level data, and to demonstrate the usefulness of data-driven analysis for a small business.

The analysis is based on primary data collected from the handwritten bookkeeping records maintained by the store, covering a one-year period from 7 February 2024 to 31 January 2025: 2,378 sales transactions with a total revenue of ₹12,19,396.

Descriptive statistical and frequency analysis reveal considerable variation in transaction size, reflecting the coexistence of low-value retail sales and high-value wholesale sales. Bangles account for a major share of annual revenue and gross profit, driven largely by wholesale transactions, while wool products show strong seasonal dependence, particularly during winter months. Seasonal analysis indicates that the two product categories complement each other, helping maintain sales continuity throughout the year.

These insights provide a structured understanding of demand patterns, revenue drivers and profitability dynamics. They help identify the root causes of seasonal fluctuations, revenue concentration and operational inefficiencies, so that appropriate actions can be taken to improve inventory planning, manage seasonal demand and balance revenue streams.

2 Proof of Originality of the Data

Organization Details:

- Organization Name: Naina Kangan and Wool Store
- Owner’s Name: Mr. Sonit Kumar
- Address: New Market, Doiwala, Dehradun (248140), Uttarakhand

To validate the originality of the primary data used in this study, relevant documents and media were provided in the original submission, including a letter from the organization, photographs of the store, and a video interaction with the store owner. The video is primarily in Hindi and an English transcript was provided. The Google Drive and dataset links are omitted from this public copy and are available in the original submission.

3 Metadata

The data used in this project was collected from the handwritten bookkeeping records of Naina Kangan and Wool Store, which are maintained on a daily basis by the owner. Sales transactions are recorded at the time of purchase in handwritten format, including product details, quantity sold, pricing and payment status. Individual customer details are not recorded. The business operates a hybrid B2B and B2C model:

sales of Bangles Box, Bangles Set, Fancy Bangles Box and Metal Bangles are recorded as wholesale (B2B), while Bangles Box (Plus size) and all wool and accessory sales are recorded as retail (B2C).

The digitized sheet contains 2,424 rows: 2,378 sales transactions and 46 closed days with no sales. The closed days are kept to preserve the continuity of the calendar but are not counted as transactions. Each sales row corresponds to a single sale on a specific date. The data runs from 7 February 2024 to 31 January 2025, capturing a complete annual business cycle, which was chosen because sales depend strongly on seasons and festivals across both wholesale and retail segments. A sample of the data is shown below.

Metadata structure:

- Date: date on which the transaction occurred
- Month: month corresponding to the transaction date
- Product Category: broad category of the product (Bangles and Wool)
- Product Type: specific product (Bangles Box, Bangles Box (Plus size), Fancy Bangles Box, Metal Bangles, Bangles Set, Wool Ball (Quality 1), Wool Ball (Quality 2), Wool Ball Packet, Baby Soft Wool, Knitting Needle (Pair) and Button)
- Quantity and Unit Type: quantity sold and its unit (box, yarn, packet, jar or pieces)
- Price per Unit and Cost per Unit: selling price and cost price per unit
- Total Revenue: total sales value of the transaction
- Gross Profit: total revenue minus cost of goods sold
- Season: season during which the transaction took place
- Customer Type: B2B or B2C
- Payment Mode: cash or online

Date	Product Type	Qty	Unit	Price	Cost	Revenue	Gross Profit	Cust.
07-02-2024	Baby Soft Wool	3	Jar	200	160	600	120	B2C
07-02-2024	Bangles Box	3	Box	130	100	390	90	B2B
07-02-2024	Bangles Box (Plus size)	3	Box	150	100	450	150	B2C
07-02-2024	Wool Ball(q2)	2	Yarn	280	240	560	80	B2C
07-02-2024	Wool Ball(q2)	2	Yarn	280	240	560	80	B2C
08-02-2024	Wool Ball Packet	1	Packet	300	265	300	35	B2C
08-02-2024	Baby Soft Wool	1	Jar	200	160	200	40	B2C
08-02-2024	Wool Ball(q2)	3	Yarn	280	240	840	120	B2C
08-02-2024	Wool Ball(q1)	1	Yarn	260	220	260	40	B2C
09-02-2024	Bangles Box	3	Box	130	100	390	90	B2B
09-02-2024	Wool Ball(q2)	2	Yarn	280	240	560	80	B2C
09-02-2024	Wool Ball Packet	2	Packet	300	265	600	70	B2C

Figure 1: Sample Portion of Sales Data (first 12 transactions)

4 Descriptive Statistics

This section presents the descriptive statistical results of the transaction-level sales data. It summarizes sales volume, revenue and profitability across time and product categories using aggregate and average measures.

Monthly Sales Volume, Revenue, and Profit Summary

Month	Quantity (box/pieces/pound)	Total Revenue (₹)	Gross Profit (₹)
January	716.5	83,160	14,553.00
February	476.5	83,396	14,990.00
March	647	91,038	16,574.00
April	643	73,755	16,300.00
May	492	57,415	13,972.00
June	508	62,415	15,472.00
July	1,959.5	2,09,570	51,285.00
August	892	86,950	20,973.00
September	240.5	24,990	5,579.00
October	3,038.5	3,19,897	73,554.00
November	509	55,078	10,227.50
December	650.75	71,732	12,082.50
Grand Total	10,773.25	12,19,396	2,65,562.00

Table 1: Monthly Aggregate Sales Performance

The monthly figures show noticeable differences in both volume and revenue, with business activity concentrated in a limited number of months and lower activity in off-season periods. October and July emerge as peak months in both sales volume and revenue, indicating strong festive demand for bangles and seasonal wholesale buying; wool adds to winter revenue. September records the lowest sales and profit, reflecting off-season conditions. Gross profit closely follows revenue, suggesting a relatively stable cost structure across months.

Average Transaction-Level Statistics by Month

Month	Avg Quantity	Avg Total Revenue (₹)	Avg Gross Profit (₹)
January	2.63	305.74	53.50
February	2.84	496.40	89.23
March	3.46	486.83	88.63
April	4.47	512.19	113.19
May	5.07	591.91	144.04
June	4.62	567.41	140.65
July	7.20	770.48	188.55
August	6.91	674.03	162.58
September	3.82	396.67	88.56
October	6.65	699.99	160.95
November	2.56	276.77	51.39
December	2.32	256.19	43.15
Overall	4.53	512.78	111.67

Table 2: Monthly Average Transaction Statistics (per sales transaction)

Average per-transaction values capture the typical size and value of a transaction in each month. Higher averages in July and October indicate festival-driven wholesale purchases, mainly of bangles. Lower averages in November and December suggest a higher proportion of small retail (B2C) wool sales.

Product Category-wise Descriptive Statistics

Product Category	Quantity	Total Revenue (₹)	Gross Profit (₹)
Bangles	8,574	8,97,620	2,16,920
Wool	2,199.25	3,21,776	48,642
Grand Total	10,773.25	12,19,396	2,65,562

Table 3: Product Category-wise Aggregate Sales Performance

Revenue and profit differ substantially between the two categories. Bangles contribute 73.6% of revenue and 81.7% of gross profit; wool contributes 26.4% of revenue and 18.3% of gross profit.

Measures of Variability

Variable	Minimum	Maximum	Standard Deviation
Quantity Sold	0.25	28.00	4.56
Total Revenue (₹)	30.00	3,250.00	416.04
Gross Profit (₹)	8.00	840.00	104.20

Table 4: Measures of Variability for Key Numerical Variables

The wide range between minimum and maximum values reflects the coexistence of small retail transactions and large wholesale bangle purchases. The standard deviations indicate notable dispersion in transaction size, consistent with a business serving both B2B and B2C customers across products with varied price points and units.

Frequency Analysis

Product Category	Frequency	Percentage (%)
Bangles	1,415	59.50
Wool	963	40.50
Total	2,378	100.00

Table 5: Frequency Distribution by Product Category

Bangles account for a higher proportion of total transactions than wool products.

5 Detailed Explanation of Analysis Process/Method

This section describes the systematic approach used to analyze the sales data of Naina Kangan and Wool Store, designed to convert manually recorded transactions into meaningful business insights while maintaining accuracy, consistency and relevance to the study objectives.

The primary sales data was maintained by the owner in handwritten registers and was first entered into a structured Excel sheet, taking care to record original values without modification. Data cleaning ensured consistency, including standardization of product names and unit types and validation of calculated fields such as total revenue and gross profit. Days with no transactions (holidays) were retained in a separate list to preserve calendar continuity and are not counted as transactions. No individual customer details

were recorded because the business does not maintain customer-level records; B2B and B2C transactions were recorded together.

The dataset variables were grouped into time-related, product-related, financial and transaction-related categories: dates and seasons; product information; price, cost, revenue and gross profit; and customer type and payment information.

A descriptive statistical approach was the primary technique. Measures of central tendency and dispersion were calculated for quantity sold, total revenue and gross profit to summarize transaction behavior, with minimum, maximum and standard deviation used to understand fluctuations in transaction size and profitability, particularly given small retail wool purchases alongside bulk wholesale bangle purchases. Frequency and percentage analysis examined the distribution of transaction counts.

Given the seasonal nature of the business, time-based analysis was conducted monthly and seasonally, calculating total quantity, revenue and gross profit for each period. The dataset was further segmented by product category and customer type to compare transaction frequency, volume and profitability between wool and bangles and between B2B and B2C customers.

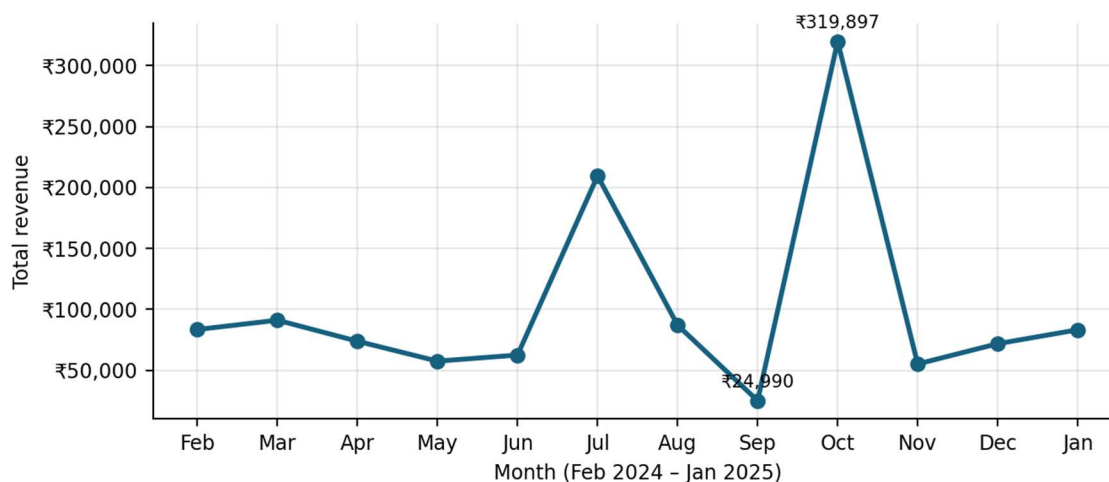
Microsoft Excel was the primary tool, used for data processing, calculation and aggregation through standard formulas and pivot tables, chosen for its effectiveness with manually digitized data. The methodology was structured to match the study objectives so that insights are logically derived from the data and give a dependable basis for the results below.

6 Results and Findings

This section presents the key findings from the transaction-level sales data. The results are supported by graphs and focus on trends, patterns and behavioral insights related to sales performance, seasonality, product categories and customer types.

Monthly Sales and Revenue Trends

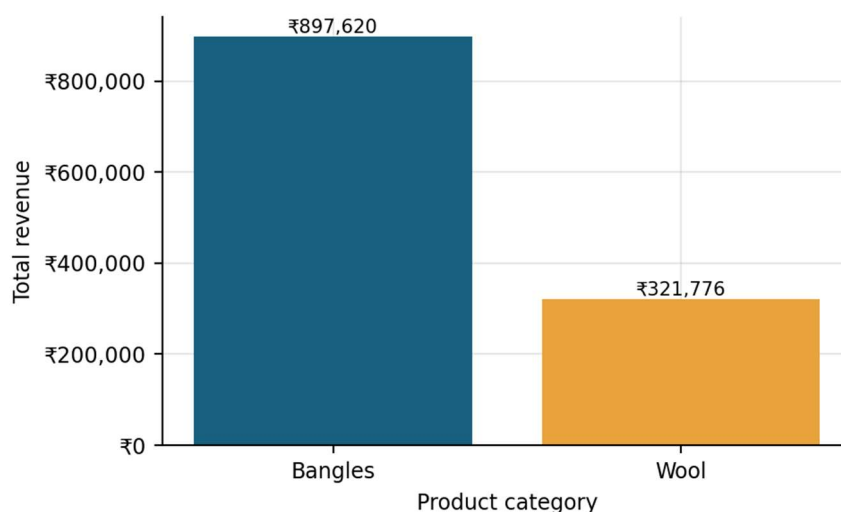
The sales transactions from 7 February 2024 to 31 January 2025 show an uneven distribution of sales over time. The line chart below shows monthly revenue, making peak months, seasonal influences and overall stability easy to identify. It also separates periods of bulk purchases from periods of frequent smaller transactions.



Graph 1: Total Revenue per Month

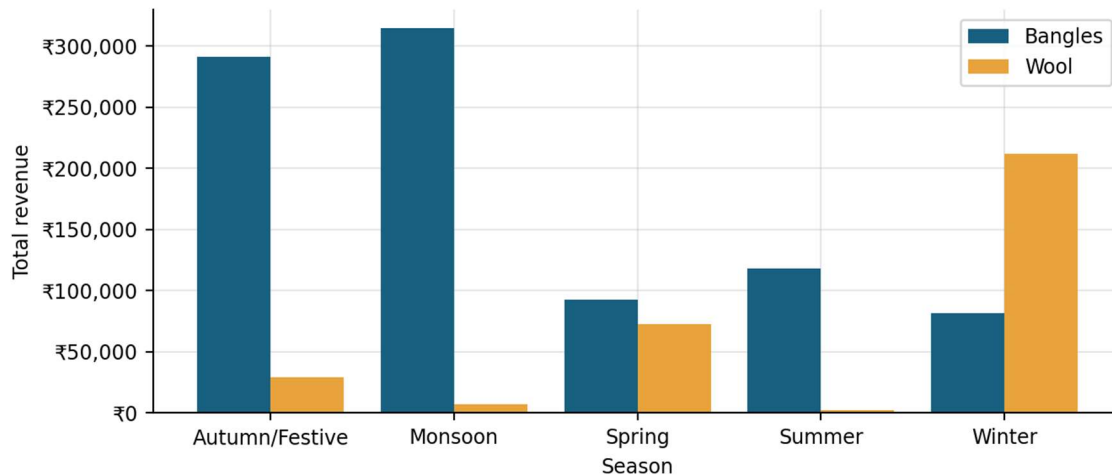
Product Category Performance

Bangles account for a larger share of total transactions, indicating high-frequency sales. This reflects steady wholesale (B2B) demand, with activity rising in festival and wedding seasons. Bangles also contribute the largest portion of revenue and gross profit, establishing them as the primary revenue-generating category.



Graph 2: Bangles vs Wool Sales over the Year

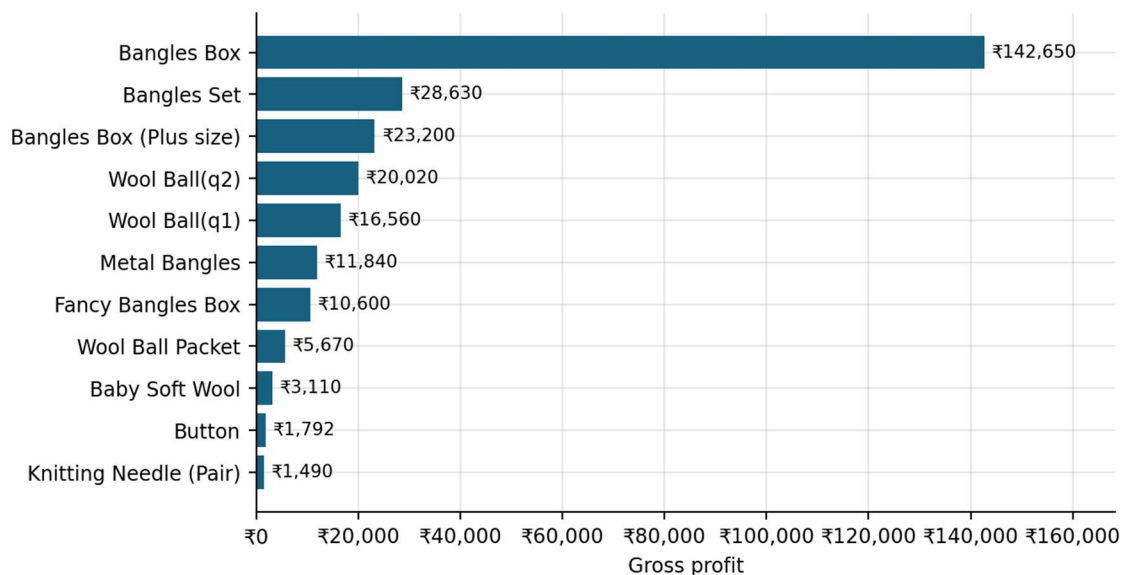
Graph 2 shows that bangles dominate annual revenue. Graph 3, which adds seasonal variation, reveals a more nuanced pattern: bangles maintain higher sales in the Autumn/Festive and Monsoon periods and decline in winter, while wool sales rise significantly in winter. This seasonal shift lets wool compensate for the decline in bangle sales and balances revenue across the year, supporting continuity of sales and cash flow despite fluctuations in individual product demand.



Graph 3: Bangles vs Wool Sales Across Different Seasons

Product-type wise Profit Contribution

The gross profit distribution by product type (Graph 4) shows a concentrated profit structure. Bangles Box is the single largest contributor, with ₹1,42,650 (53.7% of gross profit). Wool products such as Wool Ball (q1), Wool Ball (q2) and Baby Soft Wool contribute smaller but consistent shares. Although individually small, their collective contribution becomes significant in the winter season. This concentration highlights the importance of high-volume bangle products for profitability, while the several smaller wool products diversify profit sources and reduce reliance on a single product type.



Graph 4: Distribution of Gross Profit Across Product Types